

Newton Raphson Method

This method is generally used to improve the result obtained by one of the previous methods. Let x_0 be an approximate root of $f(x) = 0$ and let $x_1 = x_0 + h$ be the correct root so that $f(x_1) = 0$. Expanding $f(x_0 + h)$ by Taylor's series, we obtain

$$f(x_0) + hf'(x_0) + \frac{h^2}{2!}f''(x_0) + \dots = 0$$

Neglecting the second and higher-order derivatives, we have

$$f(x_0) + hf'(x_0) = 0$$

which gives

$$h = -\frac{f(x_0)}{f'(x_0)}$$

A better approximation than x_0 is, therefore, given by x_1 , where

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Successive approximations are given by $x_2, x_3, \dots, x_{n+1}$, where

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

which is the Newton-Raphson formula.

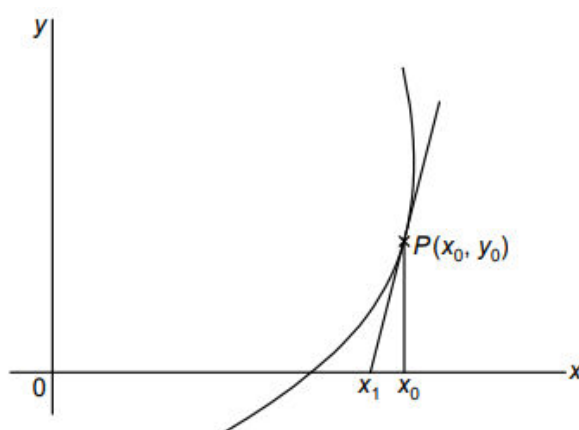


Fig. Newton-Raphson Method

Example: Use the Newton-Raphson method to find a root of the equation $x^3 - 2x - 5 = 0$.

Solution: Here $f(x) = x^3 - 2x - 5$ and $f'(x) = 3x^2 - 2$.

$$x_{n+1} = x_n - \frac{x_n^3 - 2x_n - 5}{3x_n^2 - 2}$$

Choosing $x_0 = 2$, we obtain $f(x_0) = -1$ and $f'(x_0) = 10$. Putting $n = 0$ in above equation, we obtain

$$x_1 = 2 - \left(-\frac{1}{10}\right) = 2.1$$

Now,

$$f(x_1) = (2.1)^3 - 2(2.1) - 5 = 0.061$$

And,

$$f'(x_1) = 3(2.1)^2 - 2 = 11.23$$

Hence

$$x_2 = 2.1 - \frac{0.061}{11.23} = 2.094568$$

Example: Find a real root of the equation $x = e^x$, using the Newton Raphson method.

Solution: We write the equation in the form

$$f(x) = xe^x - 1 = 0$$

Let $x_0 = 1$. Then

$$x_1 = 1 - \frac{e - 1}{2e} = \frac{1}{2} \left(1 + \frac{1}{e}\right) = 0.6839397$$

Now $f(x_1) = 0.3553424$, and $f'(x_1) = 3.337012$,

so that $x_2 = 0.6839397 - \frac{0.3553424}{3.337012} = 0.5774545$

Proceeding in this way, we obtain

$$x_3 = 0.5672297 \text{ and } x_4 = 0.5671433.$$